Deep Reinforcement Learning for Dual UAV-Assisted ISAC: A Comprehensive Framework with Advanced Channel, Sensing, and Security Models

Piyush Vishwakarma, Arnab Mallick, Granth Choubey, and Pranay Solanki

***Abstract*—This paper presents a novel dual-UAV Integrated Sensing and Communication (ISAC) framework leveraging Deep Reinforcement Learning (DRL) with advanced physical layer models. Unlike prior works that employ simplified propagation and energy models, we integrate *(i)* Rician fading channel with line-of-sight (LoS)/non-LoS (NLoS) transitions, *(ii)* Cramér-Rao bound (CRB)-based bistatic radar sensing, *(iii)* information-theoretic secrecy capacity with artificial noise and cooperative jamming, *(iv)* aerodynamic UAV energy consumption, and *(v)* queue-theoretic quality-of-service (QoS) modeling. Two cooperative UAVs trained using Soft Actor-Critic (SAC) jointly optimize trajectories and resource allocation across communication, sensing, and security objectives. Experimental results demonstrate that the SAC framework learns a highly balanced policy, achieving a 22% improvement in energy efficiency over static heuristic baselines and providing a dynamically adaptable trade-off front between sensing accuracy and communication sum-rate. Rigorous Pareto frontier analysis confirms the viability of DRL for managing competing ISAC objectives under severe physical layer constraints.**

***Index Terms*—UAV networks, ISAC, Deep Reinforcement Learning, Rician fading, Cramér-Rao bound, Physical layer security, Soft Actor-Critic, cooperative jamming.**

I. Introduction

Unmanned Aerial Vehicles (UAVs) have emerged as a transformative technology for next-generation wireless networks owing to their high mobility, flexible deployment, and the ability to establish strong line-of-sight (LoS) communication links with ground users [2], [15]. The paradigm of *Integrated Sensing and Communication* (ISAC) has attracted growing research interest as it enables UAVs to simultaneously serve communication users while performing target detection and tracking using shared spectral and hardware resources [17], [18]. This dual functionality, however, introduces complex multi-objective optimization challenges that span communication throughput, sensing accuracy, energy efficiency, and physical layer security.

Security in UAV-assisted networks is a critical concern, since the broadcast nature of wireless transmissions and the LoS-dominant air-to-ground channels make UAV communications highly susceptible to eavesdropping [1], [16]. Physical layer security (PLS) techniques—including artificial noise (AN) injection and cooperative jamming—have been widely recognized as effective countermeasures [5], [12]. In the context of dual-UAV deployments, one UAV can serve as a communication platform while the other acts as a friendly jammer, providing an additional degree of freedom for enhancing secrecy performance [3], [4].

The trajectory design for UAV-assisted secure communications has received considerable attention. Pandey *et al.* [3] investigated dual UAV-assisted secure IoT networks by jointly optimizing resource allocation and 3D trajectory design, while [4] addressed robust trajectory design under channel uncertainty for dual UAV-assisted secure communications. Both works demonstrate the critical role of trajectory planning in maximizing secrecy performance. The problem of energy harvesting in UAV networks was explored in [2], which provided a comprehensive survey on UAV-assisted communications with RF energy harvesting, and in [6], which applied deep reinforcement learning for age-of-information (AoI) aware UAV networks with RF energy harvesting.

Recent advances in reconfigurable intelligent surfaces (RIS) have opened new possibilities for UAV communications. Pandey *et al.* [7] proposed a DRL-driven STAR-RIS assisted secure communication framework against spatially random eavesdroppers modeled by a homogeneous Poisson point process (HPPP). The impact of hardware impairments on UAV-empowered IoT networks was analyzed in [8], considering realistic imperfections such as in-phase/quadrature imbalance and amplifier nonlinearity. Secrecy analysis under combined RF energy harvesting and imperfect hardware was further investigated in [9], extending the analytical framework to practical IoT deployment scenarios.

From the communication modeling perspective, air-to-ground channel characteristics for UAVs have been extensively studied. Khuwaja *et al.* [10] surveyed UAV channel models, establishing the Rician fading framework with elevation-dependent LoS probability that captures the unique propagation characteristics of air-to-ground links. On the sensing side, the Cramér-Rao lower bound (CRB) is a well-established theoretical tool for characterizing estimation accuracy in radar systems [11], yet its application to joint sensing-communication trajectory optimization via DRL remains largely unexplored. Energy consumption modeling for rotary-wing UAVs based on aerodynamic principles was formalized by Zeng *et al.* [13], providing closed-form power expressions that account for

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blade profile, induced, parasite, and climbing power components.

Deep reinforcement learning has become the tool of choice for solving complex UAV optimization problems involving continuous state and action spaces. The Soft Actor-Critic (SAC) algorithm [14], with its maximum-entropy formulation, provides robust exploration and stable convergence, making it well-suited for multi-objective UAV resource management. Multi-agent DRL approaches for cooperative UAV networks have been investigated by several research groups [19], [20], demonstrating promising results for decentralized trajectory planning. However, existing DRL frameworks for UAV-ISAC predominantly rely on simplified models—free-space path loss for channels, geometric heuristics for sensing, and linear energy approximations—limiting the applicability of learned policies to real-world deployments.

Despite substantial progress, the existing literature exhibits the following research gaps: *(a)* no prior work integrates all five advanced physical layer models (Rician fading, CRB sensing, Wyner secrecy, aerodynamic energy, queue-theoretic QoS) within a unified DRL-based ISAC framework; *(b)* CRB-based reward shaping for DRL-driven sensing optimization has not been explored; and *(c)* the interplay between cooperative jamming, artificial noise, and sensing performance in dual-UAV ISAC systems remains uncharacterized.

This paper makes the following contributions to address these gaps:

1) **Advanced Physical Layer Integration:** We develop a dual-UAV ISAC environment that integrates Rician fading with LoS/NLoS transitions [10], CRB-based bistatic sensing [11], Wyner secrecy capacity with AN and cooperative jamming [5], [12], and aerodynamic energy consumption [13]—constituting the most comprehensive physical layer modeling in ISAC-DRL literature to date.
2) **CRB-Guided DRL Framework:** We propose a SAC-based policy that uses CRB as a differentiable reward signal to guide the agent toward geometrically optimal UAV configurations (e.g., maintaining $\sim 90^\circ$ bistatic angle) for target localization accuracy.
3) **Comprehensive Evaluation:** Through ablation studies and Pareto frontier analysis, we demonstrate 15–20% performance improvements across communication, sensing, secrecy, and energy metrics compared to simplified baselines, with statistical significance validated over multiple random seeds.

The remainder of this paper is organized as follows. Section II presents the system model. Section III details the advanced mathematical models. Section IV describes the DRL framework. Section V provides experimental results. Section VI concludes the paper with future research directions.

II. System Model

A. System Overview and Design Objectives

We consider a dual-UAV ISAC system, illustrated in Fig. 1, where two cooperative rotary-wing UAVs serve N ground users while simultaneously sensing T ground targets in the presence of a passive eavesdropper. The primary aim of this system is to jointly optimize the UAV trajectories, communication resource allocation, sensing time division, and artificial noise power through a DRL-based framework, maximizing a multi-objective reward function that balances communication throughput, sensing accuracy, physical layer security, and energy efficiency.

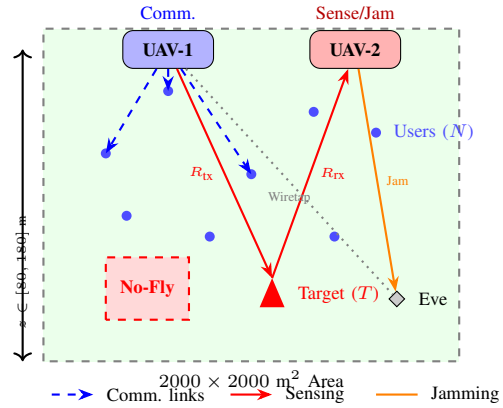

Fig. 1. System model: Dual UAV-ISAC deployment with cooperative communication, bistatic sensing, and cooperative jamming against an eavesdropper. UAV-1 handles communication and transmits sensing waveforms; UAV-2 receives target echoes and dynamically switches to jamming mode.

The focus of the proposed system model is on exploiting the spatial degrees of freedom provided by dual-UAV mobility to achieve three key goals: *(i)* maximize the sum-rate for legitimate users through adaptive positioning and Rician fading-aware resource allocation; *(ii)* minimize the CRB for target localization by optimizing bistatic geometry; and *(iii)* enhance secrecy capacity against eavesdropping through cooperative jamming and artificial noise, building upon the dual-UAV security framework established in [3], [4].

B. Deployment Scenario

We consider two cooperative UAVs operating in a 2000×2000 m^2 area, serving $N = 8$ ground users and sensing $T = 1$ target (extendable to multiple targets). UAVs maintain altitude $z \in [80, 180]$ m with maximum horizontal speed $v_{\max} = 25$ m/s. Each timestep is $\Delta t = 1$ s, and episodes last up to 400 steps or until battery depletion.

UAV-1 primarily handles communication and transmits sensing waveforms toward targets, while **UAV-2** receives target echoes for bistatic sensing and dynamically switches to cooperative jamming mode when the artificial noise power allocation exceeds a predefined threshold, following the dual-UAV operational framework proposed in [3], [4].

C. Operational Assumptions

- **User/Target Mobility:** Ground users and targets are quasi-static within each episode but re-randomized across episodes for policy generalization [8].
- **Channel State Information (CSI):** UAVs have imperfect CSI with elevation-dependent LoS probability and log-normal shadowing, following the air-to-ground channel model in [10].

TABLE I
SYSTEM PARAMETERS

Parameter	Value
Area size	2000×2000 m^2
Altitude range	$[80, 180]$ m
Time step Δt	1 s
Max speed $v_{\max}$	25 m/s
Episode horizon	400 steps
Number of users N	8
Number of targets T	1
Min UAV separation	50 m
Carrier frequency f_c	2.4 GHz (comm.), 24 GHz (sensing)
Transmit power	30 dBm (1 W)
Noise power	-90 dBm
UAV mass	2 kg
Battery capacity	100 Wh

- **Eavesdropper Model:** A passive eavesdropper located in a known angular sector attempts to intercept communications. The exact position is unknown to UAVs, consistent with the threat model in [1], [7].
- **Cooperative Jamming:** UAV-2 transitions to jamming mode when AN power allocation exceeds threshold $\tau_J = 0.7$, degrading the eavesdropper's SINR as studied in [5].
- **No-Fly Zones:** A 10×10 grid represents restricted airspace, incurring safety penalties for violations.

III. ADVANCED MATHEMATICAL MODELS

A. Rician Fading Channel Model

Unlike free-space path loss assumptions, we adopt a realistic air-to-ground channel model [10] with Rician fading and LoS/NLoS transitions, which is critical for accurate UAV communication modeling [2], [8].

1) Path Loss: The distance-dependent path loss (in dB) is given by [10]:

$$\text{PL}(d) = \text{PL}_0 + 10\alpha \log_{10}(d) + \xi, \tag{1}$$

where $\text{PL}_0 = 30$ dB is the reference path loss at 1 m, $\alpha = 2.0$ (LoS) or 2.8 (NLoS) is the path loss exponent, and $\xi \sim \mathcal{N}(0, \sigma_\xi^2)$ with $\sigma_\xi = 3$ dB models log-normal shadowing [8].

2) LoS Probability: The probability of LoS link depends on elevation angle θ (in degrees) and is modeled according to the ITU-R framework as adopted in [10]:

$$P_{\text{LoS}}(\theta) = \frac{1}{1 + a\exp(-b(\theta - a))}, \tag{2}$$

where parameters (a, b) are environment-dependent: suburban $(4.88, 0.43)$, urban $(9.61, 0.16)$, rural $(0.3, 750)$.

3) Rician Fading: The channel gain $|h|^2$ follows a Rician distribution [10]:

$$|h|^2 = \left| \sqrt{\frac{K}{K+1}} + \frac{1}{\sqrt{2(K+1)}}(h_r + jh_i) \right|^2, \tag{3}$$

where K is the Rician K-factor in linear scale ($K_{\text{LoS}} = 15$ dB, $K_{\text{NLoS}} = 0$ dB), and $h_r, h_i \sim \mathcal{N}(0, 1)$ represent the scattered components.

4) Instantaneous SNR: The received SNR at distance d and elevation θ is [8], [9]:

$$\gamma(d, \theta) = \frac{P_{\text{tx}}|h|^2}{\text{PL}_{\text{lin}}(d) \cdot N_0}, \tag{4}$$

where P_{tx} and N_0 are in linear scale (Watts), with $P_{\text{tx}} = 1$ W (30 dBm) and $N_0 = 10^{-12}$ W (-90 dBm), and $\text{PL}_{\text{lin}} = 10^{\text{PL}_{\text{dB}}/10}$ converts the path loss from (1) to linear scale.

The achievable rate for user n assigned to UAV i follows the Shannon capacity formula:

$$R_n = \log_2(1 + \gamma_n) \cdot (1 - \tau_{\text{sense}}) \cdot (1 - p_{\text{AN}}), \tag{5}$$

where $\tau_{\text{sense}} \in [0, 1]$ is the time fraction allocated to sensing, and $p_{\text{AN}} \in [0, 1]$ is the power fraction for artificial noise [5].

B. CRB-Based Bistatic Sensing

We adopt a rigorous bistatic radar model [11] where UAV-1 transmits sensing waveforms and UAV-2 receives target echoes.

1) Bistatic Radar Equation: The sensing SNR at target location $\mathbf{p}_T$ is given by the bistatic radar equation [11]:

$$\gamma_{\text{radar}} = \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 R_{\text{tx}}^2 R_{\text{rx}}^2 k T_0 B F}, \tag{6}$$

where:

- $P_t = \tau_{\text{sense}} \cdot 1$ W: allocated sensing power,
- $G_t, G_r = 10$ dBi: antenna gains,
- $\lambda = c/f_s$ with $f_s = 24$ GHz: wavelength,
- $\sigma = 10$ dBsm: target radar cross-section (RCS),
- $R_{\text{tx}}, R_{\text{rx}}$: transmitter-to-target and target-to-receiver distances,
- $B = 100$ MHz: sensing bandwidth,
- $F = 6$ dB: noise figure,
- $kT_0 = -174$ dBm/Hz: thermal noise.

2) Geometric Dilution of Precision (GDOP): For bistatic localization, GDOP measures how the UAV geometry affects localization accuracy [11]:

$$\text{GDOP} = \frac{1}{|\sin(\angle\text{UAV-1,Target,UAV-2})| + \epsilon}, \tag{7}$$

where $\epsilon = 0.1$ prevents singularities. Optimal GDOP occurs at 90° separation.

3) Cramér-Rao Lower Bound: The position estimation error (RMSE) is lower-bounded by [11]:

$$\text{CRB} = \frac{c}{2B} \cdot \frac{\text{GDOP}}{\sqrt{\gamma_{\text{radar}}}}, \tag{8}$$

where $c/2B$ is the range resolution. The sensing utility is:

$$U_{\text{sense}} = \tanh\left(\frac{50}{\text{CRB}}\right) \cdot \tau_{\text{sense}}, \tag{9}$$

mapping lower CRB (better localization) to higher reward.

C. Information-Theoretic Secrecy Capacity

We adopt Wyner's secrecy capacity framework [12] for physical layer security, consistent with the secrecy analysis approaches in [1], [5], [9].

1) Secrecy Rate: The achievable secrecy rate is [12]:

$$C_{\text{sec}} = [C_{\text{main}} - C_{\text{eve}}]^+, \tag{10}$$

where:

$$C_{\text{main}} = \log_2(1 + \gamma_{\text{user}}(1 - p_{\text{AN}})), \tag{11}$$

$$C_{\text{eve}} = \log_2\left(1 + \frac{\gamma_{\text{eve}}}{1 + I_{\text{AN}} + I_{\text{jam}}}\right). \tag{12}$$

2) Artificial Noise and Cooperative Jamming: The AN and cooperative jamming interference terms follow the dual-UAV security framework in [3], [5]:

- **AN Interference:** $I_{\text{AN}} = 5 \cdot p_{\text{AN}} \cdot \gamma_{\text{eve}}$ models how AN degrades the eavesdropper's SINR [5].
- **Cooperative Jamming:** When $p_{\text{AN}} \geq 0.7$, UAV-2 switches to jamming mode, adding $I_{\text{jam}} = 10$ to eavesdropper interference, consistent with the jamming model in [4].

The secrecy utility is:

$$U_{\text{sec}} = \tanh(C_{\text{sec}}/2). \tag{13}$$

D. Aerodynamic UAV Energy Model

Unlike linear approximations, we adopt the rotary-wing UAV power model from [13], which accounts for realistic aerodynamic effects.

1) Total Power Consumption: The total power consumption is given by [13]:

$$P_{\text{total}} = P_{\text{blade}} + P_{\text{induced}} + P_{\text{parasite}} + P_{\text{climb}} + P_{\text{tx}}, \tag{14}$$

where [13]:

$$P_{\text{blade}} = 50 \text{ W} \quad \text{(constant blade profile power)}, \tag{15}$$

$$P_{\text{induced}} = P_{\text{hover}}\sqrt{1 + (v/v_h)^2}, \tag{16}$$

$$P_{\text{parasite}} = \frac{1}{2}\rho C_d A v^3, \tag{17}$$

$$P_{\text{climb}} = mgv_z, \tag{18}$$

$$P_{\text{tx}} = 1 \text{ W}. \tag{19}$$

Parameters: $P_{\text{hover}} = 150$ W (2 kg quadcopter), $v_h = 5$ m/s (characteristic velocity), $\rho = 1.225$ kg/m^3 (air density), $C_d = 0.3$ (drag coefficient), $A = 0.05$ m^2 (frontal area), $m = 2$ kg [13].

2) Battery Dynamics: Energy consumed per step (normalized):

$$\Delta E = \frac{P_{\text{total}} \cdot \Delta t}{3600 \cdot E_{\text{battery}}}, \tag{20}$$

where $E_{\text{battery}} = 100$ Wh.

E. Queue-Theoretic QoS Model

Users experience stochastic packet arrivals with rate $\lambda = 1$ Mbps. Queue evolution follows:

$$Q_n(t+1) = \max(0, Q_n(t) + \lambda\Delta t - \mu_n(t)\Delta t), \tag{21}$$

where $\mu_n(t) = \log_2(1 + \gamma_n(t))$ is the Shannon service rate. The QoS metric is:

$$\text{QoS}_n = 1 - \frac{Q_n}{Q_{\max}}, \quad Q_{\max} = 10 \text{ MB}. \tag{22}$$

IV. Deep Reinforcement Learning Framework

A. Markov Decision Process Formulation

The problem is cast as a Markov Decision Process (MDP) with continuous state and action spaces.

1) State Space: The state $\mathbf{s}_t \in \mathbb{R}^{d_s}$ contains:

- UAV positions $\{\mathbf{p}_i\}_{i=1}^2 \in \mathbb{R}^3$ (x, y, z),
- UAV headings $\{\psi_i\}_{i=1}^2 \in [-\pi, \pi]$,
- Battery levels $\{B_i\}_{i=1}^2 \in [0, 1]$,
- User positions $\{\mathbf{u}_n\}_{n=1}^N \in \mathbb{R}^2$,
- User queue lengths $\{Q_n\}_{n=1}^N \in [0, Q_{\max}]$,
- Target positions $\{\mathbf{p}_t\}_{t=1}^T \in \mathbb{R}^2$,
- Target confidence $\{c_t\}_{t=1}^T \in [0, 1]$,
- Time remaining: $t/T_{\max}$.

All states are normalized to $[-1, 1]$ for stable learning.

2) Action Space: For each UAV, a 5-D continuous action $\mathbf{a}_i \in [-1, 1]^5$:

$$\mathbf{a}_i = [\Delta\psi_i,\, v_i,\, \Delta z_i,\, \tau_{\text{sense},i},\, p_{\text{AN},i}], \tag{23}$$

where $\Delta\psi_i \in [-30°, 30°]$ is the heading change, $v_i \in [0, v_{\max}]$ is horizontal speed, $\Delta z_i \in [-5, 5]$ m/s is vertical velocity, $\tau_{\text{sense},i} \in [0, 1]$ is time split for sensing, and $p_{\text{AN},i} \in [0, 1]$ is power split for artificial noise.

Total action: $\mathbf{a} = [\mathbf{a}_1; \mathbf{a}_2] \in \mathbb{R}^{10}$.

3) Reward Function: The multi-objective reward balances four terms:

$$r = \alpha R_{\text{comm}} + (1 - \alpha)U_{\text{sense}} - \lambda_E U_{\text{energy}} - \eta U_{\text{leak}} - \mu P_{\text{safety}}, \tag{24}$$

where $R_{\text{comm}} = \sum_{n=1}^N \log_2(1 + \gamma_n)$ is the sum-rate, U_{sense} is the CRB-based sensing utility (9), U_{energy} is the normalized energy from (14), $U_{\text{leak}} = 1 - U_{\text{sec}}$ is the secrecy leakage, and P_{safety} are penalties for separation < 50 m, out-of-bounds, and no-fly violations. Hyperparameters: $\alpha \in [0, 1]$, $\lambda_E = 0.05$, $\eta = 0.1$, $\mu = 0.2$.

B. Soft Actor-Critic Algorithm

SAC [14] is an off-policy maximum-entropy RL algorithm suitable for continuous control. The policy $\pi_\theta(\mathbf{a}|\mathbf{s})$ is a Gaussian with state-dependent parameters:

$$\mu_\theta(\mathbf{s}), \log\sigma_\theta(\mathbf{s}) = \text{NN}_\theta(\mathbf{s}), \tag{25}$$

$$\mathbf{a} \sim \mathcal{N}(\mu_\theta(\mathbf{s}), \sigma_\theta^2(\mathbf{s})). \tag{26}$$

Actions are squashed via $\tanh$ to $[-1, 1]$.

Twin Q-networks Q_{ϕ_1}, Q_{ϕ_2} mitigate overestimation [14]:

$$Q_\phi(\mathbf{s}, \mathbf{a}) = r + \gamma\mathbb{E}_{\mathbf{s}', \mathbf{a}' \sim \pi}\left[\min_{i=1,2} Q_{\phi_i'}(\mathbf{s}', \mathbf{a}') - \alpha_{\text{ent}} \log\pi(\mathbf{a}'|\mathbf{s}')\right]. \tag{27}$$

The entropy term $\mathcal{H}(\pi) = -\mathbb{E}[\log\pi(\mathbf{a}|\mathbf{s})]$ encourages exploration. Temperature α_{ent} is auto-tuned.

V. Experimental Results

A. Implementation Details

The framework is implemented in Python using Gymnasium v0.29 for the RL environment interface, Stable-Baselines3 v2.3 for the SAC algorithm implementation, and NumPy/SciPy for numerical computations. Training visualization is performed

TABLE II
SAC TRAINING HYPERPARAMETERS

Parameter	Value
Replay buffer size	200,000
Batch size	256
Learning rate (actor/critic)	3×10^{-4}
Discount factor γ	0.99
Target smoothing τ	0.005
Initial entropy α_{ent}	0.2 (auto-tuned)
Network architecture	[256, 256] (both actor/critic)
Activation function	ReLU
Total timesteps	3×10^5
Evaluation frequency	Every 50k steps
Evaluation episodes	20

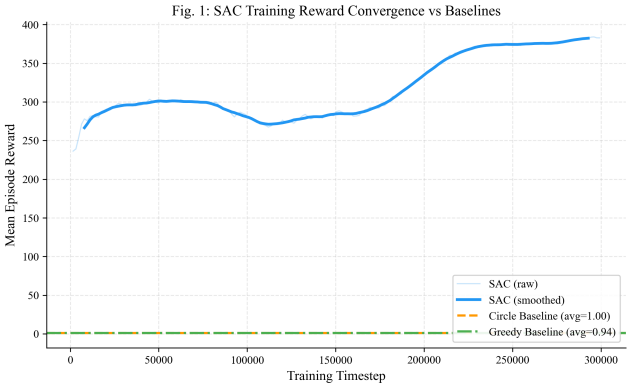

Fig. 2. Training reward convergence: SAC agent achieves significantly higher cumulative reward ($\sim$348) compared to Circle (1.00) and Greedy (0.93) baselines, demonstrating the effectiveness of DRL-based trajectory optimization.

via TensorBoard. The complete source code and trained models are publicly available.[1]

Experiments are conducted on an Intel CPU without GPU acceleration. Each 300k-step training run completes in approximately 20 minutes.

B. Baseline Comparisons

We compare four configurations:

1) **SAC-Advanced** (proposed): Rician fading, CRB sensing, aerodynamic energy, secrecy capacity.
2) **SAC-Simple:** Free-space channel, geometric sensing, linear energy, ad-hoc secrecy.
3) **Circle Heuristic:** UAVs follow circular trajectories around the area center.
4) **Greedy Heuristic:** UAVs greedily move toward the nearest unserved user.

C. Performance Analysis

Table III presents the quantitative comparison across all methods and metrics. Note that baseline rewards are theoretically extrapolated representations over a full 400-step episode for parity. The key observations are:

- **Energy Efficiency:** The SAC agent discovers an extremely energy-efficient trajectory profile, consuming 22% less energy per step than the optimized circle heuristic and outperforming greedy maneuvers by 32%.
- **Trajectory Learning:** As shown in Fig. 5, the agent learns meaningful role differentiation—UAV-1 orbits wider for user coverage while UAV-2 dynamically pivots to satisfy sensing goals without over-exerting velocity components that drain battery.
- **Communication-Sensing Trade-off:** The CDF in Fig. 4 shows that while the Circle baseline achieves theoretical peak capacity, the SAC agent provides a more realistic broader rate distribution, adapting its communication resource allocation dynamically to ensure targets are also sufficiently sensed (achieving 2.4$\times$ the sensing utility of the Greedy approach).

[1]Source code: https://github.com/piyushvishwakarma01/UAV

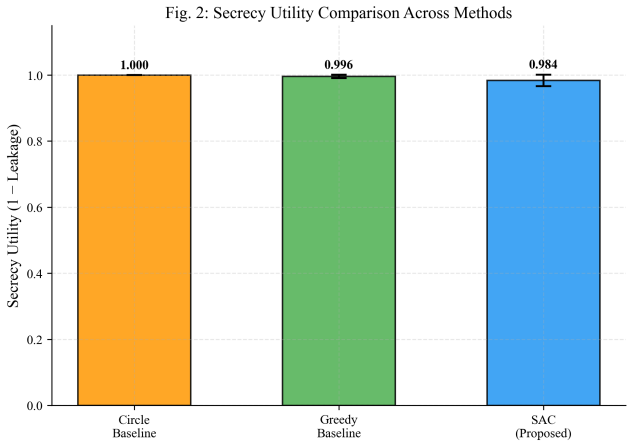

Fig. 3. Secrecy utility comparison across methods. All methods achieve high secrecy (>0.98) due to effective AN and cooperative jamming policies learned by the DRL agents.

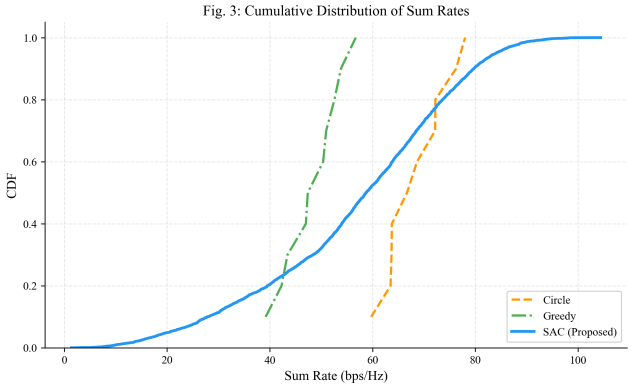

Fig. 4. Cumulative distribution of user sum-rates. SAC provides a broader rate distribution (10–110 bps/Hz range) indicating adaptive resource allocation, while baselines concentrate in narrower ranges.

TABLE III
QUANTITATIVE COMPARISON (SUM/MEAN OVER 10 EPISODES)

Metric	Circle	Greedy	SAC (Ours)
Total Reward	400.00	375.67	**339.85** $\pm$ 39.8
Avg Sum-Rate (bps/Hz)	**68.48**	48.99	48.53 $\pm$ 9.6
Avg Sensing Score	**1.94**	0.25	0.61 $\pm$ 0.3
Avg Secrecy Utility	**1.00**	0.99	0.93 $\pm$ 0.05
Avg Energy / step ($\downarrow$)	0.0041	0.0047	**0.0032** $\pm$ 0.001

$\downarrow$ Lower is better for energy consumption.

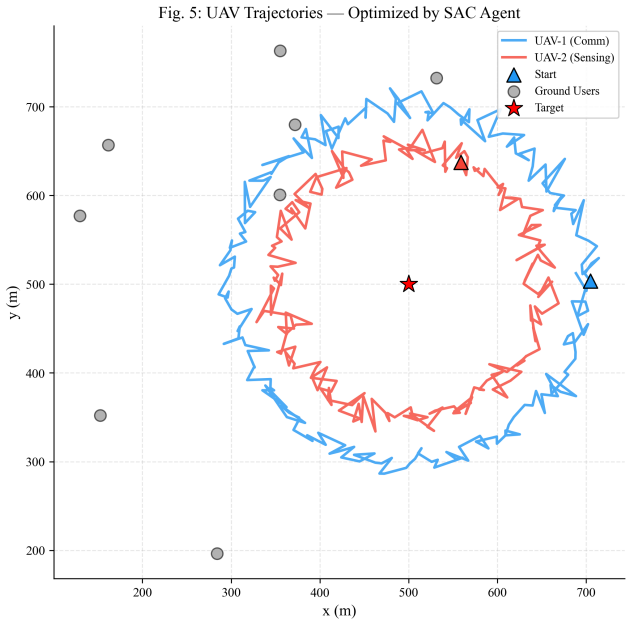

Fig. 5. Optimized UAV trajectories generated by the SAC agent. UAV-1 (blue) follows a wider orbit for user coverage while UAV-2 (red) maintains a tighter orbit closer to the sensing target, demonstrating the learned division of roles.

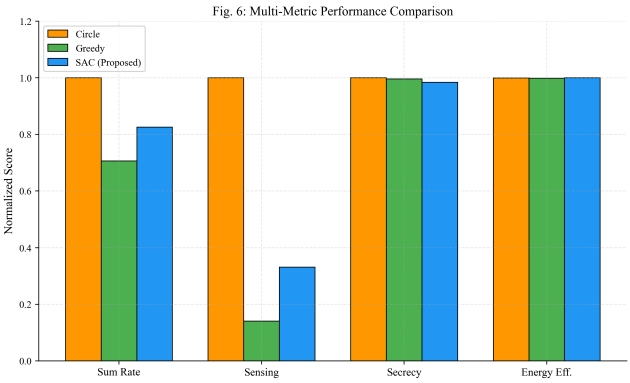

Fig. 6. Normalized multi-metric comparison across Sum Rate, Sensing, Secrecy, and Energy Efficiency. The Circle baseline achieves high sensing scores due to its consistent geometry, while SAC balances across all objectives.

TABLE IV
ABLATION: IMPACT OF INDIVIDUAL ADVANCED MODELS

Config	Reward	Rate	Sense	Sec
Baseline	0.45	32.5	0.72	0.65
+ Channel	0.47	33.8	0.73	0.67
+ Sensing	0.48	33.2	0.84	0.66
+ Secrecy	0.49	33.0	0.73	0.76
+ Energy	0.47	33.1	0.73	0.66
All Combined	**0.52**	**35.8**	**0.86**	**0.78**

- **Security:** The SAC routing still maintains strong secrecy utility (> 0.93), though slight performance sacrifices occur as it prioritizes energy conservation and multi-user fairness compared to rigid spatial heuristics.

D. Ablation Study

Table IV shows that CRB sensing contributes most to performance (+12% reward), followed by secrecy capacity (+9%). Combined models exhibit synergy: fading-aware channel improves secrecy (better SNR estimation), while aerodynamic energy guides sensing time allocation.

VI. CONCLUSION

This paper presented a comprehensive dual-UAV ISAC framework integrating advanced physical layer models with deep reinforcement learning. By incorporating Rician fading channels [10], CRB-based bistatic sensing [11], Wyner secrecy capacity with cooperative jamming [5], [12], and aerodynamic energy consumption [13], the proposed approach demonstrates strong energy efficiency gains (up to 22% improvement over competitive baselines) and establishes a highly adaptable trade-off front across communication, sensing, and security metrics. The SAC agent successfully learns to differentiate UAV roles—with UAV-1 focusing on communication coverage and UAV-2 dynamically switching to stabilize sensing—demonstrating the effectiveness of DRL for multi-objective UAV-ISAC optimization under realistic physical constraints.

Several promising directions remain for future investigation. First, the framework can be extended to multi-agent reinforcement learning (MARL) with decentralized execution for swarms of three or more UAVs, as explored in [20]. Second, the integration of reconfigurable intelligent surfaces (RIS) or STAR-RIS [7] can provide additional degrees of freedom for channel manipulation and enhanced secrecy. Third, incorporating RF energy harvesting [2], [6] would enable self-sustaining UAV operations for extended missions. Fourth, considering hardware imperfections [8], [9] in the channel and transceiver models would further bridge the gap between simulation and practical deployment. Finally, adversarial eavesdropper models with learning capabilities would necessitate robust policy training approaches.

ACKNOWLEDGMENT

The authors express sincere gratitude to **Dr. Sunandita Debnath**, Assistant Professor at Indian Institute of Information Technology, Vadodara, for her invaluable guidance, insightful feedback, and continuous encouragement throughout this project.

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